

UNIVERSIDAD PANAMERICANA
CAMPUS GUADALAJARA

POSGRADOS DE INGENIERÍA



**Data recovery in 5G wireless channels with doppler and
delay effect using mixture of experts for enhanced Industry
4.0 mobility.**

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1 Abstract

Project proposes a framework for data recovery in wireless communication systems, focusing on dynamic environments characterized by Doppler effects and delay-induced distortions. The solution integrates a mixture of experts by selecting the strengths of neural networks and traditional approaches, the framework is designed to intelligently manage high-mobility scenarios such as vehicle-to-vehicle communication and Industry 4.0 applications. A classifier-based intelligent administrator, which dynamically optimizes the trade-off between time, computational resources, and performance, ensuring efficient handling of signal distortions under varying conditions.

This address challenges of real-time communication in high-mobility scenarios by providing a scalable and adaptive solution that significantly improves performance while meeting computational and time constraints.

2 Keywords

- 5G
- OFDM
- Doppler
- Mixture of experts
- Vehicle-to-vehicle communication
- High mobility Scenarios
- 802.11p
- Cognitive Performance Analysis
- IoT
- Industry 4.0
- Equalization
- Neural Networks (NN)
- Recurrent Neural Networks (RNN)
- Gated Recurrent Unit (GRU)
- Deep Neural Networks (DNN)
- Vision Transformers (ViT)

3 Introduction

Aquí va el contenido de Chapter-Introduction.

3.1 Background

Aquí va el contenido de Chapter-Background.

3.2 Theoretical Framework And State Of The Art

Aquí va el contenido de Chapter Theoretical Framework And State Of The Art.

4 Problem Statement

Aquí va el contenido de Problem Statement.

5 Justification

Industry 4.0 represents a potential business for the future, as evidenced by the substantial growth in investments in related technologies. In 2019 alone, investments in IoT in India reached approximately USD 5 billion [2]. These technologies enable real-time data collection, processing, and transmission, with applications in critical sectors such as energy and water management in developed countries [1].

The growth of the Internet of Things (IoT) is given by the increasing proliferation of connected devices and the adoption of advanced technologies such as 4G, 5G, and Wi-Fi, which rely on OFDM modulation and PSK coding to ensure communication efficiency [5]. While these technologies are widely used in daily mobile and office applications, they also face significant challenges due to adverse environmental conditions, including high temperatures, humidity, dust accumulation, and limited maintenance capacity and skilled personnel [6].

A less explored aspect of IoT is its the implementation on high-mobility devices, where equipment operates at variable speeds [3]. This variability introduces distortions in the carrier signal, such as ambiguous Doppler effects, which become significant at speeds starting from 25 m/s and distances of 500 m, with a minimum frequency shift of 5 kHz [4]. These distortions affect the accuracy of encoded and transmitted data, posing challenges for applications requiring high reliability.

6 Chapter-Objectives

Aquí va el contenido de Chapter-Objectives.

7 Hypothesis

7.1 Investigation

The Doppler effect and delay in high-mobility wireless communication channels generate distortions in the signal-to-noise ratio and inter-signal interference, negatively affecting the recovery of transmitted data through 5G and OFDM networks.

7.2 Null

The Doppler effect and delay in high-mobility wireless communication channels do NOT generate significant distortions in the signal-to-noise ratio or inter-signal interference.

7.3 Alternative

The distortions in the signal-to-noise ratio and inter-signal interference in high-mobility channels are NOT primarily caused by the Doppler effect and delay but rather by other factors such as thermal noise, external interference, or modulation errors.

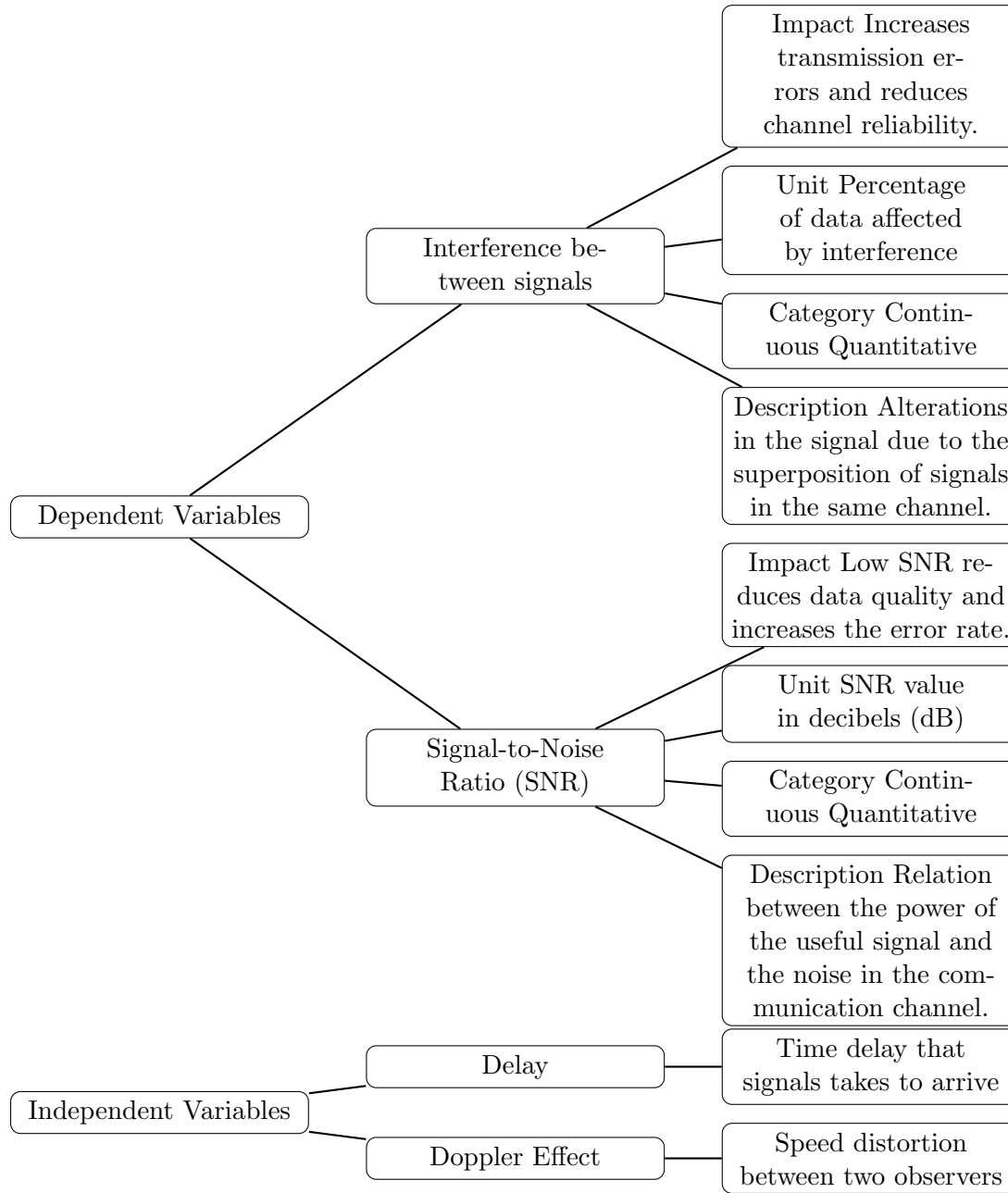
8 Methodology

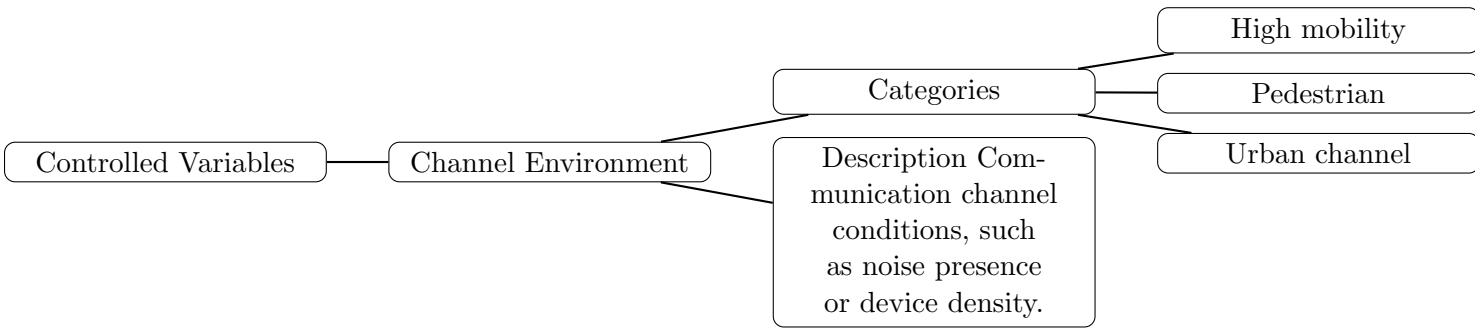
Articles

Author	Title	Source	Year	Link	Main Topic	Technique Model Used	Relevant Results
Sneha Chennamsetty	Beyond Traditional Techniques: Utilizing Neural Networks to Improve and Compare the Performance of OFDM and OTFS	Research Square	2024	https://doi.org/10.2196/rs.3.rs-5017954/v1	Mejoras de Modulacion	NN, RNN, GRU, LSTM	RNN supero a otros modelos y optimizo OFDM y OTFS, mejorando la comunicacion en canales dinámicos.
Sneha Chennamsetty	Analysis of Neural Networks Based OTFS Wireless System	IEEE Xplore	2023	https://ieeexplore.ieee.org/document/1043112	Mejoramiento de la comunicacion	NN, RNN, DNN	The Neural networks (NN) model has the ability to perform multiple tasks at once
Sneha Chennamsetty	Cognitive Performance Analysis of the Proposed Testbed for OFDMA Wireless System	IEEE Xplore	2024	https://ieeexplore.ieee.org/document/1043112	The legit. proposed testbed consists of software-defined radios (SDRs) that enable the dynamic configuration of radio parameters and the emulation of realistic wireless environments	SDR, NN	Testbeds as learning platform tools for simulating the 4G and 5G cellular networks in real time

Author	Title	Source	Year	Link	Main Topic	Technique Model Used	Relevant Results
Sneha Chennam- setty	Analysis of PAPR in OTFS Modulation with Classical Selected Map- ping Technique	IEEE Xplore	2023	https://ieeexplore.ieee.org/document/10041313	high data rate, better flexibility, requires a less cyclic prefix and less amount of peak to aver- age power ratio (PAPR)	OTFS	This paper presents the PAPR anal- ysis in OTFS modulation by using classical SLM and the ob- tained results are compared against the OFDM mod- ulation.
Sneha Chennam- setty	Analysis and emulation of proposed testbed for OFDMA 4G and 5G cellular networks	IEEE Xplore	2024	https://onlinelibrary.wiley.com/doi/10.1002/real.5749	Real-time analysis for 4G and OFDM Time signals.	4G and 5G Real Time	4G and 5G OFDMA cellular networks in real time
Wei Wang	Rep ViT: Revisiting Mobile CNN From ViT Perspective	IEEE Xplore	2023	https://ieeexplore.ieee.org/document/10359437	Vision/Computa- cional	Rep ViT	Adaptacion de CNN en disposi- tivos moviles con perspectiva ViT para rendimiento superior y eficien- cia energética.
Wei Wang	L2A: Learning Affinity from Attention for Weakly Supervised Continual Semantic Segmentation	IEEE Xplore	2023	https://ieeexplore.ieee.org/document/10246896	Segmentation Semántica	Aprendizaje de Afinidad con Aten- cion	Mejoras en segmentacion débilmente super- visada mediante el aprendizaje de afinidades y atencion.

Author	Title	Source	Year	Link	Main Topic	Technique Model Used	Relevant Results
Wei Wang	MVEB: Self-Supervised Learning With Multi-View Entropy Bottleneck	IEEE Xplore	2020	https://ieeexplore.ieee.org/document/9095223	Aprendizaje No Supervisado	Entropía de Bottleneck Multi-View	Propone el uso de un Bottleneck para mejorar el aprendizaje auto-supervisado mediante múltiples perspectivas.





9 Research Details

Aquí va el contenido de Chapter Research Details.

10 Preliminary Results

Aquí va el contenido de Chapter Preliminary Results.

11 Chapter-Conclusions

Aquí va el contenido de Chapter-Conclusions.

Bibliography

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